

Damir Kumukov

Prof. Alraee

Data 205

25 March 2026

For my project I will be using two datasets that focus on electric vehicle registration and infrastructure. The first dataset is the MVA Electric and Plug-in Hybrid Vehicle Registrations by County, which gives monthly counts of electric vehicle (EV) registrations across Maryland from 2020-2025. This dataset is very useful for analyzing how EV adoption has increased over time and how it varies across different counties. The second dataset I will be using is the Alternative Fuel Stations dataset from the U.S Department of Energy. This dataset gives me information on EV charging station location, charger type, and if it's private or public.

One challenge with these datasets is that the EV count dataset had historical data, while the charging station dataset is more of a current snapshot for February 17, 2026. Also, the charging station dataset might need cleaning, especially to get county level information. Despite these challenges, both datasets can be merged at the county level to compare EV adoption with infrastructure availability.

Goals and Research Questions

My main goal from this project is to determine whether charging infrastructure in Maryland is expanding at a pace that matches the growth of electric vehicle adoption. To support this goal, I will explore 8 questions;

1. Which counties have the highest EV adoption?

Method: I will group data by county then get the sum of EV registrations and sort from highest to lowest to show the findings on a bar chart and also make a heatmap to show EV density.

2. Which counties have the most charging stations?

Method: Filter dataset to Maryland + electric stations then count the number of stations per county, rank counties and plot bar chart and add a GIS map to show station locations

3. How has EV adoption changed over time?

Method: Group EV data by month or year then calculate the total EVs per time period and plot a line chart

4. Why do some MD counties have so much EV adoptions compared to other counties

Method: Compare high EV counties vs low EV counties by looking at EV counts and charging station counts. I also plan on doing online research

5. Are there counties with high EV adoption but relatively few charging stations?

Method: Create a dataset with EV counts and station counts by county, compare values side by side, identify counties with high EV counts but low station counts and plot using a scatter plot

6. What is the EV-to-charger ratio for each county?

Method: First calculate: EV registrations divided by number of charging stations then rank counties by ratio and visualize using a color scaled bar chart and heat map

7. Do fast chargers have a stronger relationship with EV adoption than Level 2 chargers?

Method: Use multiple linear regression to compare the impact of fast chargers and Level 2 chargers on EV adoption.

Model: EV Registrations $\sim$ Fast Chargers + Level 2 Chargers

8. Does infrastructure appear to lag behind or keep up with EV growth?

Method: Compare EV growth trends with infrastructure levels, use regression model:
EV registrations $\sim$ charging stations, interpret slope and relationship using trend lines graph.

My expectation is to find that some counties, especially more urban or higher income areas, both have higher EV adoption and more charging stations, but there may also be gaps where infrastructure has not kept pace.

Methods

I will use Python in Google Colab to complete this project. The main steps will include data cleaning, merging datasets, exploratory data analysis, and building simple models. For analysis, I plan to use:

- Exploratory Data Analysis (EDA) to clean the data and find trends and patterns
- Multilinear Regression
- Visualizations like a GIS maps to show infrastructure distribution, heatmap and bar chart to compare EV adoption and charging infrastructure
- Linear regression to analyze the relationship between EV registrations and charging station availability

I will also calculate key metrics like EV to charger ratio to find whether infrastructure is keeping pace with demand. These methods fit my project because both datasets contain numerical and time based data that can be analyzed using regression and visualization techniques.

Risks and Mitigations

A major risk is the mismatch in time coverage between the two datasets. The EV dataset includes multiple years of data, while the charging dataset has a more recent snapshot. To make the datasets work

together I will focus on relative comparisons, such as EV to charger ratios and general trends, rather than relying entirely on time series modeling. Another risk is data inconsistency, such as missing values or differences in formatting between datasets. I will fix this by cleaning the data, standardizing column formats, and removing incomplete records if needed. A third risk is merging datasets because of differences in geographic identifiers. If this becomes a problem, I will aggregate both datasets at the county level to ensure consistency. As a backup plan, I can still analyze each dataset separately and compare results conceptually.

Milestone Timeline

Project Timeline EV Adoption

